

LRRK2 phosphorylation of Rab GTPases in Parkinson's disease

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Rab GTPases comprise a large family of conserved GTPases that are critical regulators of the secretory and endocytic pathways. The human genome encodes ~ 65 Rabs that localize to discrete membrane compartments and, when in their GTP-bound state, bind to effector proteins to carry out diverse functions. Activating mutations in LRRK2 kinase cause Parkinson's disease, and subsets of Rab GTPases are important LRRK2 substrates. LRRK2 phosphorylates a conserved threonine residue that is essential for Rab interaction with guanine nucleotide exchange factors, effectors, and GDI that recycles Rabs between membrane compartments. This brief review will highlight new findings related to LRRK2-mediated phosphorylation of Rab GTPases and its consequences. Remarkably, Rab phosphorylation flips a switch on Rab effector selection with dominant consequences for cell pathophysiology.

Keywords: GTPase; Parkinson's disease; primary cilia; protein phosphorylation

Introduction to LRRK2 and its pathophysiological functions

Pathogenic, activating mutations in the Leucine Rich Repeat Kinase 2 (LRRK2) cause Parkinson's disease [1,2], thus there is great interest in identifying the substrates of this kinase. In 2016, the teams of Matthias Mann and Dario Alessi reported a tour de force phospho-proteomic analysis carried out to identify the precise substrates of LRRK2, the kinase that is mutated in inherited Parkinson's disease [3]. One might imagine that a simple comparison of phospho-peptides obtained from cells with and without a specific kinase inhibitor would be sufficient for this purpose, but kinase inhibitors are not adequately specific and can have significant off-target effects. In a first-round screen, Steger et al. [3] used two structurally distinct LRRK2 kinase inhibitors, hoping that the converging 76 hits would decrease the identification of off-target

substrates. They then carried out a second screen that utilized a third LRRK2 inhibitor (MLi-2), in conjunction with a cell line expressing an LRRK2 protein engineered such that it could not bind MLi-2 (LRRK2 A2016T). In this case, any differences seen would have been due to the specific loss of LRRK2 inhibitor-binding capacity rather than the activity of another, off-target kinase. This second screen yielded 128 phosphosites that were specifically downregulated in the engineered cells. Remarkably, the overlap of the data obtained from these separate screens led to the joint identification of only a single hit: threonine 73 (T73)-phosphorylated Rab10 protein.

Several other Rab GTPases were also later shown to be specific substrates; these additional GTPases may not have appeared in the above-mentioned screens due

Abbreviations

ANK, ankyrin; ARM, armadillo; COR, C-terminal of ROC; LRR, leucine-rich repeat; LRRK2, Leucine Rich Repeat Kinase 2; ROC, Ras of complex proteins.

to peculiarities in enzyme cleavage, physical properties of the peptides in terms of their chromatographic behaviors or ionization, or overlap with other dominant species that could mask and therefore prevent their identification. The results of both screens continue to be valuable as they may contain additional, disease-relevant hits. Importantly, Rab10 emerged from this work as a highly significant LRRK2 substrate that has been subsequently validated by many independent approaches in many different cell types and tissues.

Alignment of the entire Rab GTPase family revealed that a serine or threonine residue at position 73 is conserved in 42 Rab family members, but many fewer Rabs were shown to be phosphorylated by LRRK2. Thus, Rab8 and Rab12 were confirmed as LRRK2 substrates; later work confirmed that 14 Rabs could be phosphorylated upon LRRK2 overexpression, but just 10 Rabs are endogenous LRRK2 substrates (Rab3A/B/C/D, Rab8A/B, Rab10, Rab12, Rab35, and Rab43; [4–6]). Interestingly, Steger et al. [3] detected co-immunoprecipitation between LRRK2 and Rab8A and Rab10 proteins. Later work showed that this interaction likely reflects a mechanism by which LRRK2 is recruited to membranes (see below).

In terms of substrate specificity, in mouse embryonic fibroblasts, Rab10 is the preferred substrate of LRRK2, followed by Rab8 and Rab35, but substrate selectivity depends in part on the relative abundance of Rab proteins and the counter-action specificity and abundance of phosphatases. LRRK2-preferred substrates vary significantly between mouse tissues [7]. In MEFs, per cell Rab levels range from ~ 1.25 million copies for Rab10, ~ 0.65 million copies for Rab8A and Rab1B, ~ 0.2 million copies for Rab8b, Rab12 and Rab35, ~ 70 000 copies for Rab3A, Rab3B, Rab3D, and Rab43 and 20000 copies for Rab29,

whereas LRRK2 is of lower abundance (~ 6000 copies [7]).

Rab10's T73 lies in the middle of the so-called Rab switch II region, the structure of which reflects the identity of the bound guanine nucleotide [8]. Steger et al. [3] showed that phosphorylation at this site, within this critical region blocks the ability of Rabs to bind to their recycling factor, GDI; Rab8 phosphorylation blocks interaction with its cognate Rabin8 GEF protein (Fig. 1), and Rab binding to a long list of effectors is also blocked. Thus, phosphorylated Rabs are rendered inactive for their normal roles. Quantitative Phostag gels that can resolve phosphorylated and nonphosphorylated Rab proteins, as well as mass spectroscopy analyses, showed that only a small fraction of total Rab GTPases is phosphorylated by LRRK2 at a steady state [4,9]. This is critical because completely blocking normal Rab:effector interactions, along with Rab nucleotide exchange enzymes and GDI recycling, would be predicted to have deleterious consequences for cell physiology.

If phosphorylated Rabs cannot engage their cognate GEFs, do their effectors show the same strong preference for binding to GTP-bearing phosphorylated Rabs and not GDP-bearing phosphorylated Rabs? Waschbüsch et al. [10] showed that at least one effector, RILPL2 binds with a strong preference to GTP-bound phosphorylated Rab8A versus GDP-bound phosphorylated Rab8, typical of other Rab effector interactions. Also, in cells, a GDP-preferring Rab10 mutant was LRRK2-phosphorylated much less well than wild-type Rab10 [11]. Thus, GTP-bearing Rabs may bind effectors immediately after LRRK2 modification. A continuing mystery is why Rab phosphorylation is rapidly turned over [4]; nevertheless, such turnover provides an additional opportunity for Rabs to be acted upon by cognate GEFs before the next round of LRRK2 modification.

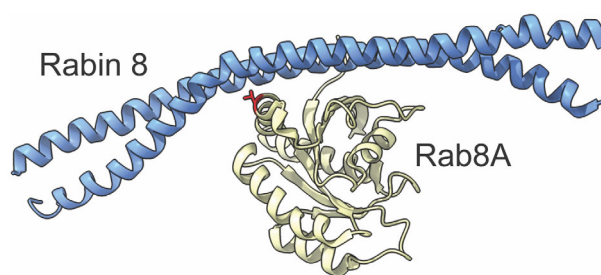


Fig. 1. Structure of the dimeric coiled coil of Rabin8 bound to Rab8A. Structure of a dimeric coiled coil from Rabin8 guanine nucleotide exchange factor (blue) bound to Rab8A GTPase (cream; PDB 4LHZ; [39]). Threonine 72 that is the target of LRRK2 phosphorylation is shown in red; note that the Thr72 side chain represents the closest point of contact between the Rab and its cognate GEF, and phosphorylation disrupts this interaction.

Membrane activation of LRRK2 kinase by Rab GTPases

Most LRRK2 is present in cytosol, but about 10% is membrane-associated where it co-localizes with substrate Rab GTPases (cf. [12]). Prior work has indicated both genetic and biochemical interactions between Rab29 and LRRK2 (see [13]), and of all the LRRK2 substrates identified to date, Rab29 is the only other substrate with a link to Parkinson's disease as the gene is located within the PARK16 locus. Rab29 is an unusual Rab GTPase in that it is poorly prenylated and, unlike most other Rab GTPases, it occurs in cytosol as a monomer instead of being bound to GDI [11]. This may be related to Rab29's much weaker affinity for guanine nucleotides than most other Rab proteins [11], which may reflect a less tightly folded protein core. But a fraction of Rab29 localizes to the Golgi complex, and when overexpressed in cells, Rab29 can recruit LRRK2 to membranes, where it becomes activated [11,12,14,15,16]. Despite Rab29's demonstrated ability to activate LRRK2 upon overexpression, Kalogeropoulou et al. [13] showed that endogenous Rab29 is not needed for intrinsic LRRK2 activation, or for the hyperactivation that is seen upon the addition of a number of lysosomotropic agents such as nigericin, monensin, chloroquine, and lysosome-disrupting LLOMe peptide [13,17].

By whatever mechanism that Rab29 can activate LRRK2 at the Golgi, LRRK2 can also be activated

by Rab29 that has been mistargeted to the surface of mitochondria [11]. This indicates that kinase activation does not absolutely require other Golgi-associated membrane components. Mitochondrially localized Rab10 is phosphorylated as efficiently as it is modified when localized to its native membrane compartments, suggesting that Rab10 may share with Rab29 the capacity of engaging and activating LRRK2 on membranes. Interestingly, when mitochondrially anchored Rab29 recruited LRRK2 to mitochondria, phosphorylated Rab10 also accumulated on that membrane surface. This is likely due to the inability of GDI to retrieve mislocalized, phosphorylated Rab proteins that may be mistakenly delivered to mitochondria [3]. Such a mistargeting event would not normally be seen in the absence of the increased Rab phosphorylation driven by exogenous LRRK2 expression. Finally, even LRRK2 anchored at the inner leaflet of the plasma membrane could lead to phosphorylated Rab10 being trapped on that unusual membrane surface. Thus, LRRK2 activation appears to require a membrane-associated, Rab-binding partner or substrate, but the identity of the membrane is not critical [11].

Rab29 and the related Rab32 and Rab38 proteins bind to LRRK2's N-terminal Armadillo domain. McGrath et al. [18] showed that Rab32-subfamily members, Rab29, Rab32, and Rab38, interact with LRRK2 residues 1–552 in a GTP-dependent manner, with residues 386–392 being critical for this interaction (Fig. 2). Vides et al. [19] showed further that Rab29

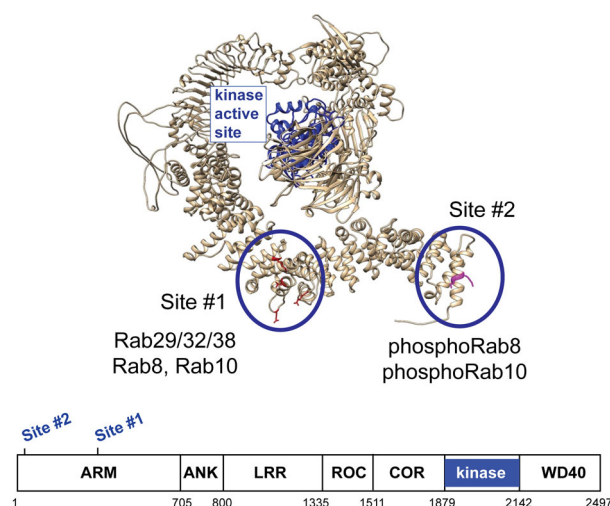


Fig. 2. Multiple Rab GTPase-binding sites in the N-terminal Armadillo domain of LRRK2. Shown is the *ALPHA*FOLD [20] predicted structure model of human LRRK2 with the active site indicated and highlighted in blue. Site #1 and site #2 of the N-terminus bind to nonphosphorylated and phosphorylated Rabs, respectively. Below the model is a linear diagram of LRRK2 domains for orientation purposes: ARM, armadillo; ANK, ankyrin; LRR, leucine-rich repeat; ROC, Ras of complex proteins; COR, C-terminal of ROC. Numbers below indicate amino acid residue number.

interacts with LRRK2's Armadillo domain in cells *via* key residues including R361, R399, L403 and K439, and that these are crucial for Rab29-mediated membrane recruitment and activation of LRRK2. Vides et al. [19] refer to this Rab29-binding site as 'Site 1' (Fig. 2) and showed that Rab8A and Rab10 likely also bind to the same site that is also detected using ALPHAFOLD [20] within CHIMERA X software [21] to identify protein–protein interaction interfaces.

Further analysis revealed a second binding site (Site 2) that is specific for phosphorylated Rab GTPases; this site includes LRRK2 K17 and K18, residues that are critical for LRRK2's membrane association in cells and onto Rab10-decorated planar lipid bilayers [19] (Fig. 2). Because the LRRK2 N-terminus binds the product of LRRK2 kinase activity, the membrane recruitment of LRRK2 is facilitated by its own kinase action as part of a feed-forward, cooperative membrane recruitment process. This will lead to a local concentration of LRRK2 in a membrane microdomain. Vides et al. [19] confirmed the cooperativity of LRRK2 membrane recruitment by reconstituting that process on planar lipid bilayers and monitoring single molecule recruitment by TIRF microscopy (Fig. 3). Recent cryoEM structures of full-length LRRK2

bound to Rab29 [22] suggest that Rab29 binding to the Armadillo domain may trigger the formation of higher-order structures that reflect the active form of the LRRK2 protein.

Phosphorylated Rab-specific effectors regulate primary cilia formation and stability

Primary cilia are microtubule-based cellular projections present on most attached cells throughout the body and are critical for developmental signaling events. They are enriched in G protein-coupled receptors and are essential for Hedgehog and Wnt signaling pathways [23]. The subset of Rab proteins that were LRRK2 substrates included several that had already been implicated in the formation of primary cilia. This triggered the Steger et al. [5] discovery that LRRK2 expression interferes with the formation of cilia in mouse embryonic fibroblasts and cultured 3T3 cells.

Careful proteomic analysis of proteins bound to endogenous or overexpressed Rab8A protein, with or without LRRK2 kinase activity, revealed a small set of proteins that seem to bind preferentially to phosphorylated Rab8 protein [5]. Most notable were RILPL1

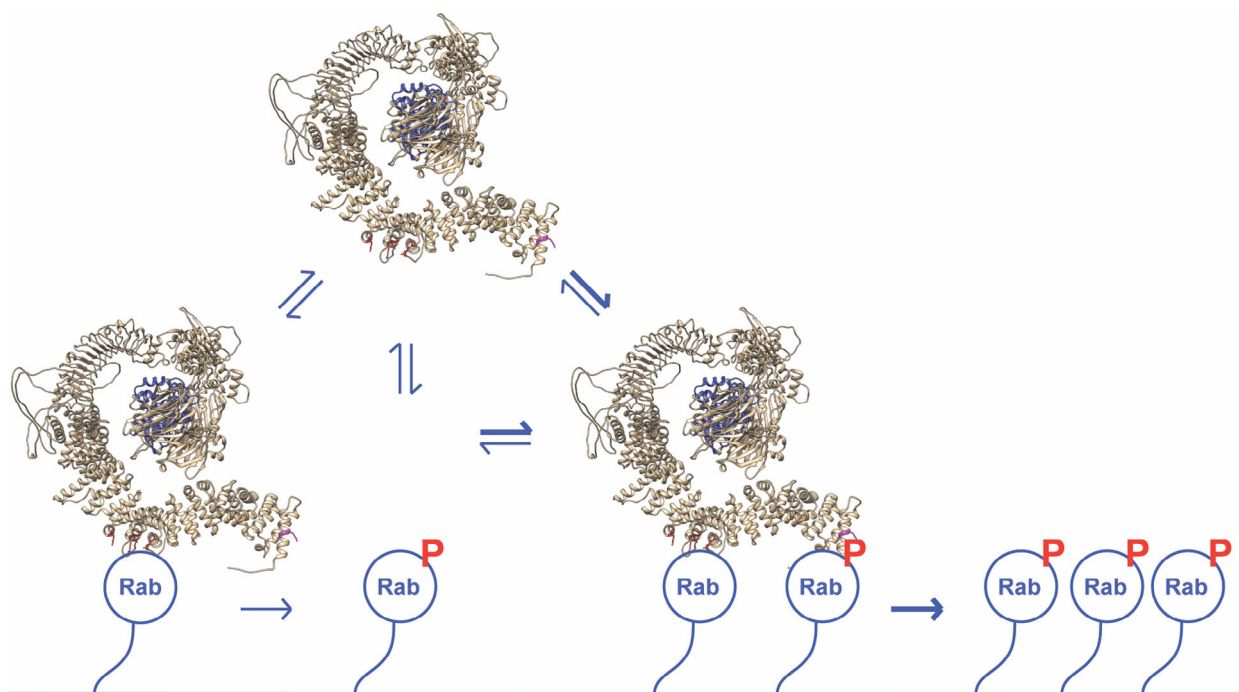


Fig. 3. Model for feed-forward, cooperative recruitment of LRRK2 onto membranes. LRRK2 can interact with nonphosphorylated Rab GTPases *via* site #1. Once membrane bound, it can generate phosphoRabs that can now engage site #2. Rab binding to both sites increases the avidity of LRRK2 for membranes and retains LRRK2 on the membrane surface to phosphorylate more Rab substrates. Modified from [19].

and RILPL2 that bind with preference to phosphorylated Rab8, phosphorylated Rab10, and phosphorylated Rab12 compared with their nonphosphorylated counterparts. RILPL1 and RILPL2 have also been shown to play some role in cilia function [24]. Thus, while Rab phosphorylation blocks most partner protein interactions, it also flips a switch for Rab effector binding and drives association with a subset of new proteins that generate dominant cellular phenotypes.

Dhekne et al. [25] went on to show that the ability of LRRK2 to block cilia formation requires Rab10 protein and RILPL1; the phosphorylated Rab10-RILPL1 complex localizes to the mother centriole in the perinuclear region of multiple cell types (see also [26]). Some phosphorylated Rab10 is still detected in the pericentriolar region in cells lacking RILPL1, suggesting additional protein partners. Loss of cilia correlates directly with a loss in Hedgehog signaling, monitored by downstream induction of Gli1 expression, in both MEF cells and in patient-derived, G2019S LRRK2 mutant iPS cells [25]. Of note, correction of the mutation restored ciliogenesis and Hedgehog signaling [25]. Phosphorylated Rab10 and RILPL1 have also been implicated in interfering with centrosome cohesion (cf. [16,27]).

Live-cell video microscopy revealed that LRRK2 activity decreases the overall probability of cilia formation in R1441C LRRK2 MEF cells; pRab10 and RILPL1 interfered with the recruitment of TTBK2 kinase that is needed to release CP110 from the mother centriole [28]. Cilia were also less stable in cells expressing LRRK2 R1441C independent of whether the cells contained Rab10 or RILPL1 proteins. Future work will be needed to define more precisely the events underlying LRRK2-mediated cilia inhibition.

RILPL2 binds to the motor protein, Myosin Va. Dhekne et al. [29] showed that Myosin Va also binds LRRK2-phosphorylated Rab10 with high affinity. RILPL2 and MyoVa move to the pericentriolar region in pathogenic LRRK2 expressing cells, by a phosphorylated Rab10-dependent process. Fluorescence loss in photobleaching microscopy showed that phosphorylated Rab10 actually retains Myosin Va over pericentriolar membranes. In RPE cells, RILPL2 is not essential for ciliogenesis, but RILPL2 overexpression blocks this process. Thus, LRRK2-generated phosphorylated Rab10 dramatically redistributes and anchors Myosin Va and RILPL2 at the mother centriole, which is likely to interfere with Myosin Va's normal role in cilia formation.

The heterotetrameric structure of phosphorylated Rab8 bound to the RILP homology (RH2) domain of RILPL2 revealed the precise residues that contribute

to the phosphorylated Rab-binding interface and the conservation of these amino acid sequences enabled Waschbüsch et al. [10] to show that JIP3 and JIP4 are also phosphorylated Rab8A- and phosphorylated Rab10-binding partners. The roles of JIP3 and JIP4 as motor protein adaptors for late endosomes and lysosomes [30–32] suggest that Rab phosphorylation may alter organelle motility by interaction with JIP3 and JIP4. Indeed, Erika Holzbauer et al. found hyperactive LRRK2 recruits JIP4 to the autophagosomal membrane, inducing abnormal activation of kinesin that could lead to an unproductive tug of war between anterograde and retrograde motors and defects in autophagosome transport and maturation [33].

Phosphomimetic and nonphosphorylatable Rab mutants are not normal

Studies of protein phosphorylation often utilize substrate mutants generated in the hope that a residue cannot be phosphorylated (T-to-A) or that a residue will mimic the phosphorylation state (T-to-E). Unfortunately, these versions of Rab8A and Rab10 are nonfunctional [25]. Nonphosphorylatable Rab8A and Rab10 mutants cannot rescue their respective ciliation phenotypes and both proteins are poorly prenylated in cells. Moreover, Rab8A-TE and Rab10-TE fail to show the enhanced interaction with RILPL1 that is characteristic of phosphorylated Rab8 and Rab10 proteins. Thus, **TA and TE mutants of Rab8A or Rab10 should not be used in future experiments exploring the consequences of Rab phosphorylation** as any resulting phenotypes will not be straightforward to interpret.

PPM1H: A Rab-specific phosphatase

Berndsen et al. [34] used anti-phosphoRab10 immunoblotting to monitor the level of Rab10 phosphorylation upon siRNA knockdown of 322 different phosphatase catalytic or regulatory subunits. A second screen sensitized the system by the addition of a LRRK2 inhibitor for a few minutes, which reveals quick phosphatase action in control cells. PPM1H phosphatase was the top hit in both types of screens; overexpression of this enzyme led to the loss of phosphoRab10. The closely related PPM1M but not PPM1J showed phosphorylated Rab10 phosphatase activity in cells, and PPM1H acted on phosphorylated Rab8A and phosphorylated Rab35 but not on phosphorylated Rab12. Purified PPM1H efficiently dephosphorylated Rab8A *in vitro*.

Berndsen et al. engineered a substrate trapping PPM1H mutant protein and elegantly showed that upon expression in cells followed by immunoprecipitation, mass spectrometry revealed bound Rab8A, Rab8B, Rab10, and Rab35, confirming that they are bona fide substrates. Finally, the knockdown of PPM1H was sufficient to decrease primary cilia formation, even in wild-type mouse embryonic fibroblasts [34]. This important finding indicates that even endogenous, wild-type LRRK2, which is present at very low copy numbers in fibroblasts, controls primary ciliogenesis.

Waschbüsch et al. [35] solved the crystal structure of a large portion of PPM1H and could show that an unusual flap domain extension provides substrate specificity for this Rab-specific enzyme. Simple insertion of PPM1H flap domain sequences in place of the related PPM1J domain was sufficient to confer Rab specificity for a normally Rab-inactive phosphatase. Future experiments will reveal the substrate specificity of PPM1J and also whether additional phosphatases dephosphorylate Rab proteins, especially Rab12 that seems to be insensitive to PPM1H action [34]. Hopefully, a co-crystal structure of PPM1H with one of its phosphorylated substrates will also reveal how this enzyme shows such high specificity for a subset of Rab GTPases.

Cilia and Parkinson's disease

How does the loss of cilia connect to Parkinson's disease? Parkinson's disease is characterized by a specific loss of dopaminergic neurons in the Substantia nigra. These neurons send highly branched processes throughout the dorsal striatum, and their firing properties are regulated by neurons in that region. Andreas Kottman and colleagues showed that dopaminergic neurons require Hedgehog for their survival, but they do not have receptors to sense Hedgehog [36,37]. Instead, they secrete Hedgehog onto rare, cholinergic interneurons in the dorsal striatum. Those neurons also require Hedgehog for viability and are thought to respond by secretion of glial-derived neurotrophic factor (GDNF) that supports dopaminergic cell viability. Dhekne et al. [25] discovered that half of the cholinergic interneurons of the dorsal striatum of LRRK2 mutant mice lose their cilia compared with their wild-type counterparts. Khan et al. [38] found this to also be true in three additional mouse models of pathogenic LRRK2 activation and discovered that half of the astrocytes of the dorsal striatum also lose their cilia. Because Hedgehog signaling is proportional to ciliation (at least in cell culture), one would predict that

these cells are less able to respond to Hedgehog signals. Indeed, Khan et al. [38] detected clear dysregulation of Hedgehog signaling in LRRK2 mutant animals associated with LRRK2 mutation and a decrease in GDNF production in cholinergic neurons from mutant animals. More work is needed to understand why a subset of these cell types loses cilia in a LRRK2 mutant background.

These experiments suggest that LRRK2 blockade of primary cilia in the astrocytes and cholinergic interneurons of the dorsal striatum decreases the ability of these cells to respond to Hedgehog signals emanating from dopaminergic neurons of the Substantia nigra. This would be predicted to contribute to the slow death of dopaminergic neurons, consistent with Parkinson's disease, a disease of old age.

Conclusion

The identification of Rab GTPases as bona fide LRRK2 substrates represents a major breakthrough in our understanding of how LRRK2 pathogenic mutations lead to Parkinson's disease. Future work in this exciting area will continue at a rapid pace to learn the full consequences of Rab GTPase phosphorylation, with hopes for benefits for Parkinson's patients.

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